Lesson Eight

Risk management

Effective Project Management: Traditional, Agile, Extreme, Hybrid 8th Edition

Ch05: What Are Project Management Process Groups?

Software Project Management 5th Edition

Ch07: Risk management

Presented by
Bowen DU

Outline

- Identify the risks that might affect the project's success.
- Assess the risks based on cost-effective principal.
- Select appropriate methods to mitigate project's risks.
- Estimate risks to the schedule using PERT & Critical

Chain technique.

Concept of Risk

- PM-BOK defines risk as *an uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives*.
- The key elements of a risk follow:
  - It relates to the future.
  - It involves "cause vs effects".
    - For example, a 'cost over-run' might be identified as a risk, but 'cost over-run' describes some damage, but does not say what causes it. .

Concept of Risk

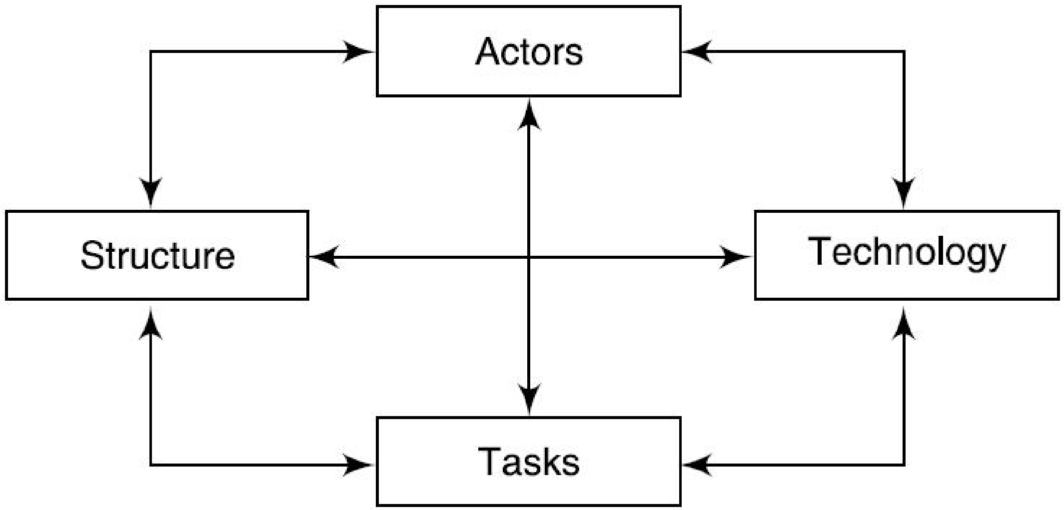
- Case: Cause vs Effects

  Match the following causes to their possible effects.

  - Causes
    - staff inexperience.
    - lack of top management commitment.
    - new technology.
    - users uncertain of their requirements.
  - Effects
    - testing takes longer than planned;
    - planned effort and time for activities exceeded;
    - project scope increases;
    - time delays in getting changes to plans agreed.

Categories of Risk

- Project risks are those that could prevent the achievement of the objectives given to the project manager & team.
- The key role of risk management is considering ***uncertainty*** remaining after a plan has been formulated.
- Sociotechnical model of risk identifies Four interlinked factors:



Risk Management Process

- Business decision is to assess how the expected loss compares to the cost of defraying(支出) all or some of the loss and then taking appropriate action.

- **risk identification**
- **risk assessment**
- **risk plan**
- **risk monitoring & control**

- What are the risks?
- What is the probability of loss that results from them?
- How much are the losses likely to cost?
- What might the losses be if the worst happens?
- What are the alternatives?
- How can the losses be reduced or eliminated?
- Will the alternatives produce other risks?

Risk Management – Risk Identification

- Technical risks
- Project management risks
- Organizational risks
- External risks

Risk Management – Risk Identification

- Technical risks

TABLE 7.1 Software project risks and strategies for risk reduction

Risk	Risk reduction techniques
Personnel shortfalls	Staffing with top talent; job matching; teambuilding; training and career development; early scheduling of key personnel
Unrealistic time and cost estimates	Multiple estimation techniques; design to cost; incremental development; recording and analysis of past projects; standardization of methods
Developing the wrong software functions	Improved software evaluation; formal specification methods; user surveys; prototyping; early user manuals
Developing the wrong user interface	Prototyping; task analysis; user involvement
Gold plating	Requirements scrubbing; prototyping; cost–benefit analysis; design to cost

Risk Management – Risk Identification

- Technical risks

TABLE 7.1 Software project risks and strategies for risk reduction

(Contd)

Late changes to requirements	Stringent change control procedures; high change threshold; incremental development (deferring changes)
Shortfalls in externally supplied components	Benchmarking; inspections; formal specifications; contractual agreements; quality assurance procedures and certification
Shortfalls in externally performed tasks	Quality assurance procedures; competitive design or prototyping; contract incentives
Real-time performance shortfalls	Simulation; benchmarking; prototyping; tuning; technical analysis
Development technically too difficult	Technical analysis; cost–benefit analysis; prototyping; staff training and development

Risk Management – Risk Identification

- Risk Identification Matrix Template

RISK CATEGORIES AND RISKS	SCOPE TRIANGLE ELEMENTS				
	Scope	Time	Cost	Quality	Resources
Technical					
Project Management					
Organizational					
External					

Risk Management – Risk Identification

- Case: Matrix Template

Candidate Risk Driver Worksheet

P/I LEGEND:	Very High	VH	MITIGATION LEGEND:	Yes	Y
	High	H		Monitor	M
	Medium	M		No	N
	Low	L			
	Very Low	VL			

Risk Category	Scope Triangle	Event #	Event	Y/N	Prob.	Impact	Priority	Mitigate Y/M/N
Tech	Scope	TS01	Available HW/SW technology limits scope					
		TS02	New technology does not integrate with old					
Tech	Time	TT01	Integrating technologies impacts schedule					
Tech	Cost	TC01	Unexpected need to acquire hardware					
Tech	Cost	TC02	Unexpected need to acquire software					
Tech	Quality	TQ01	Technology limits solution performance					
Tech	Res	TR01	New/unfamiliar technology					
Tech	Res	TR02	Inadequate software sizing					
Tech	Res	TR03	Inadequate hardware sizing					
Proj Mgt	Scope	PS01	Senior scope change request too significant					

Risk Management – Risk Identification

- Case: Matrix Template (cont's)

Ext	Scope	ES01	Changing competition					
Ext	Scope	ES02	Unexpected changes in policy, stds, regs					
Ext	Time	ET01	Competing priorities with vendors					
Ext	Cost	EC01	Unexpected price increases from vendor					
Ext	Res	ER01	Misunderstood contract terms					
Ext	Res	ER02	Unavailable vendor resources					
Ext	Res	ER03	Unexpected state budget cuts					
Ext	Res	ER04	Unexpected county budget cuts					
Uncl	Uncl	UU01	Intellectual property & copyright obstacles					
Uncl	Uncl	UU02	Problem/Project details not available					
Uncl	Uncl	UU03	Aversion to divulging sensitive information					
Uncl	Uncl	UU04	Access to information					
Uncl	Uncl	UU05	Demand for space exceeds available space					

Risk Management – Risk Assessment

- A common problem with risk identification is that a list of risks is potentially endless.
  - What is the probability of loss that results from them?
  - How much are the losses likely to cost?
  - What might the losses be if the worst happens?

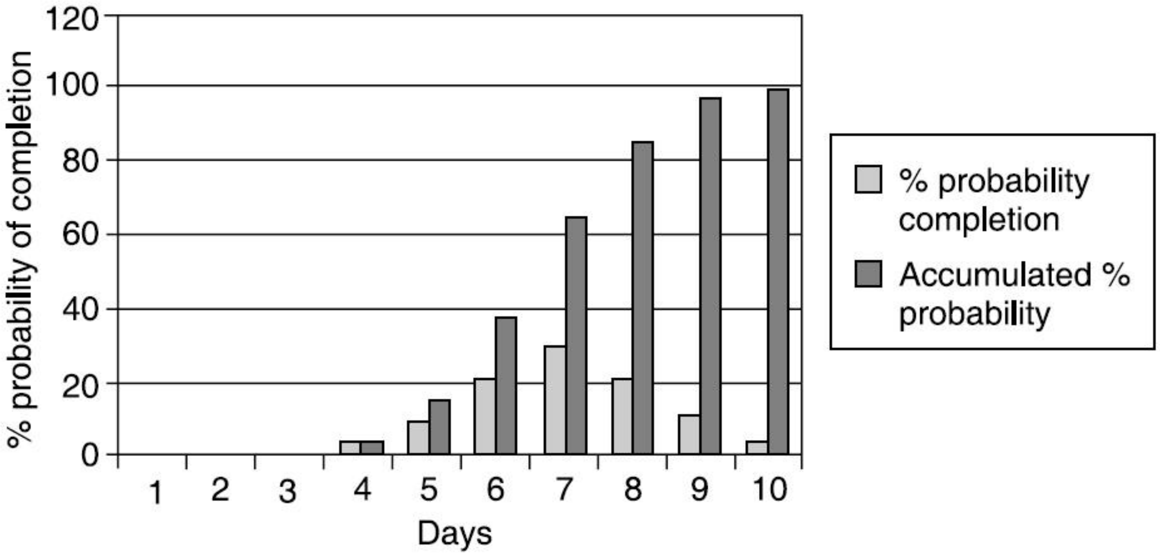
- A way is needed of distinguishing the damaging and likely risks. *Risk exposure* measures each potential risk.

risk exposure = (potential damage) × (probability of occurrence)

  - Potential damage 500k yuan when risk happens
  - Probability of occurrence is 0.1% in ten years
  - Risk exposure = 500k × 0.1% = 500 yuan

Risk Management – Risk Assessment

- Probability chart



- Barry Boehm has suggested that, because of this, both the risk losses and the probabilities be assessed using relative scales in the range 0 to 10. The two figures could then be multiplied together to get a notional risk exposure.

Risk Management – Risk Assessment

- Case of risk exposure assessment

Ref	Hazard	Likelihood	Impact	Risk
R1	Changes to requirements specification during coding	8	8	64
R2	Specification takes longer than expected	3	7	21
R3	Significant staff sickness affecting critical path activities	5	7	35
R4	Significant staff sickness affecting non-critical activities	10	3	30
R5	Module coding takes longer than expected	4	5	20
R6	Module testing demonstrates errors or deficiencies in design	4	8	32

Risk Management – Risk Assessment

- Case of risk exposure assessment

Qualitative descriptors of risk probability and associated range values

Probability level	Range
High	Greater than 50% chance of happening
Significant	30–50% chance of happening
Moderate	10–29% chance of happening
Low	Less than 10% chance of happening

Risk Management – Risk Assessment

- Case of risk exposure assessment

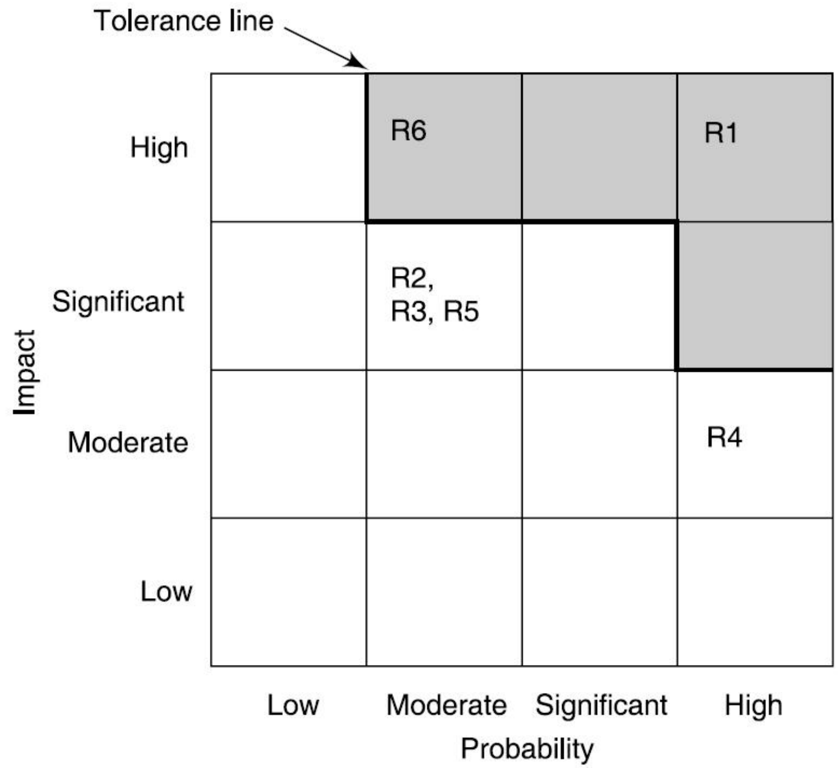
Qualitative descriptors of impact on cost and associated range values

Impact level	Range
High	More than 30% above budgeted expenditure
Significant	20 to 29% above budgeted expenditure
Moderate	10 to 19% above budgeted expenditure
Low	Within 10% of budgeted expenditure.

Risk Management – Risk Assessment

- Case of risk exposure assessment

  A probability impact matrix



Risk Management – Risk Assessment

- Case of risk exposure assessment

  A probability impact matrix table

Project Activity	A	B	C	D	E	F	G	H	I	J	Score
Rqmnts Analysis	2	3	3	2	3	3	2	2	1	1	22
Specifications	2	1	3	2	2	2	1	2	2	3	20
Preliminary Design	1	1	2	2	2	2	1	2	2	2	17
Design	2	1	2	2	2	3	1	2	2	1	18
Implement	1	2	2	3	3	2	1	2	2	1	19
Test	2	2	2	2	2	3	2	2	2	2	21
Integration	3	2	3	3	3	3	2	3	3	2	27
Checkout	1	2	2	3	3	3	2	3	2	2	23
Operation	2	2	3	3	3	3	3	3	1	1	24
Score	16	16	22	22	23	24	15	21	17	15	191

Maximum score is 270. Risk level for this project is 191/270 = 71%.

Risk Management – Risk Plan

- Having identified the major risks and allocated priorities, the task is to decide how to deal with them.
  - What are the alternatives?
  - How can the losses be reduced or eliminated?
    - Accept.
    - Avoid.
    - Reduce/Mitigate.
    - Transfer.
  - Will the alternatives produce other risks?

Risk Management – Risk Plan

- Risk accept
  - This is the do-nothing option. We will already, in the risk prioritization process, have decided to ignore some risks in order to concentrate on the more likely or damaging. We could decide that the damage inflicted by some risks would be less than the costs of action that might reduce the probability of a risk happening.
- Risk avoidance
  - Some activities may be so prone to accident that it is best to avoid them altogether. If you are worried about sharks then don't go into the water.
  - For example, given all the problems with developing software solutions from scratch, managers might decide to retain existing clerical methods, or to buy an off-the-shelf solution.

Risk Management – Risk Plan

- Risk reduction/mitigation
  - We decide to go ahead with a course of action despite the risks, but take precautions(预防措施) that reduce the probability of the risk.
  - For instance, the developers might have been promised generous bonuses to be paid on successful completion of the project.
  - Fairley's four commercial off-the-shelf software acquisition risks

Integration	Difficulties in integrating the data formats and communication protocols of different applications.
Upgrading	When the supplier upgrades the package, the package might no longer meet the users' precise requirements. Sticking with the old version could mean losing the supplier's support for the package.
No source code	If you want to enhance the system, you might not be able to do so as you do not have access to the source code.
Supplier failures or buyouts	The supplier of the application might go out of business or be bought out by a rival supplier.

Risk Management – Risk Plan

- Transfer
  - In this case the risk is transferred to another person or organization. With software projects, an example of this would be where a software development task is outsourced to an outside agency for a fixed fee. You might expect the supplier to quote a higher figure to cover the risk that the project takes longer than the ‘average’ expected time.
  - On the other hand, a well-established external organization might have productivity advantages as its developers are experienced in the type of development to be carried out.
  - The need to compete with other software development specialists would also tend to drive prices down.
  - Risk transfer is what effectively happens when you buy insurance.

Risk Management – Risk Plan

- Contingency planning
  - Risk reduction activities would appear to have only a small impact on reducing the probability of some risks, for example staff absence through illness. While some employers encourage their employees to adopt a healthy lifestyle, it remains likely that some project team members will at some point be brought down by minor illnesses such as flu. These kinds of risk need a contingency plan. This is a planned action to be carried out if the particular risk materializes. If a team member involved in urgent work were ill then the project manager might draft in another member of staff to cover that work.

Risk Management – Risk monitoring & control

- Deciding on the risk actions

Risk Reduction Leverage (RRL).

$$\text{Risk reduction leverage} = \frac{(RE_{before} - RE_{after})}{\text{cost of risk reduction}}$$

- REbefore is the risk exposure, as explained in above, before risk reduction actions have been taken. REafter is the risk exposure after taking the risk reduction action.
- Assumption: Cost-Effective Principle

Risk Management – Risk monitoring & control

- Risk Log Entrys

ID #	Risk Description	P	I	Risk Owner	Action to be Taken	Outcome

Risk Management – Risk monitoring & control

- Risk Record Template

RISK RECORD

Risk id		Risk title			
Owner		Date raised	Status		

Risk description

Impact description

Recommended risk mitigation

Probability/impact values

	Probability	Impact		
		Cost	Duration	Quality
Pre-mitigation				
Post-mitigation				

Incident/action history

Date	Incident/action	Actor	Outcome/comment

Evaluating risks to the schedule

- PERT (Program Evaluation and Review Technique)
  - PERT was developed to deal with the uncertainty surrounding estimates of task durations.
  - Suitability: high-risk, state-of-the art projects.
  - Three estimates:
    - Most likely time (m)
    - Optimistic time (a)
    - Pessimistic time (b)

  Expected duration: $t_e = (a + 4m + b)/6$

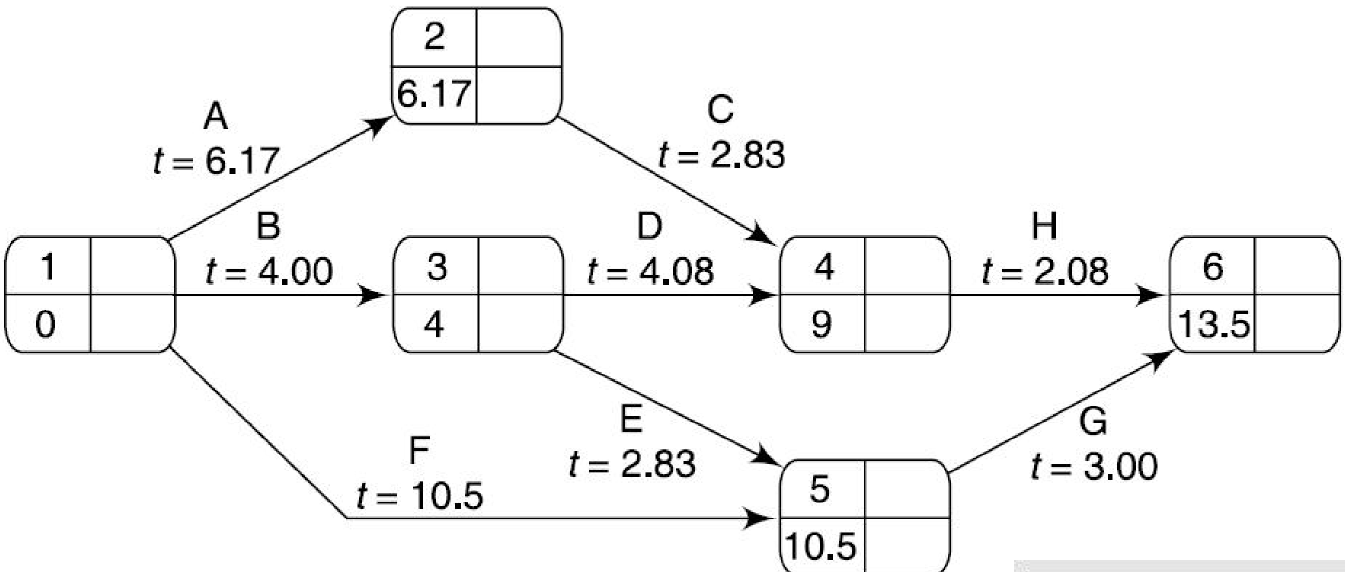
Evaluating risks to the schedule

- Case: PERT Application

Activity	Optimistic (a)	Activity durations (weeks). Most likely (m)	Pessimistic (b)
A	5	6	8
B	3	4	5
C	2	3	3
D	3.5	4	5
E	1	3	4
F	8	10	15
G	2	3	4
H	2	2	2.5

Evaluating risks to the schedule

- Case: PERT Application



Even number	Target date
Expected date	Standard deviation

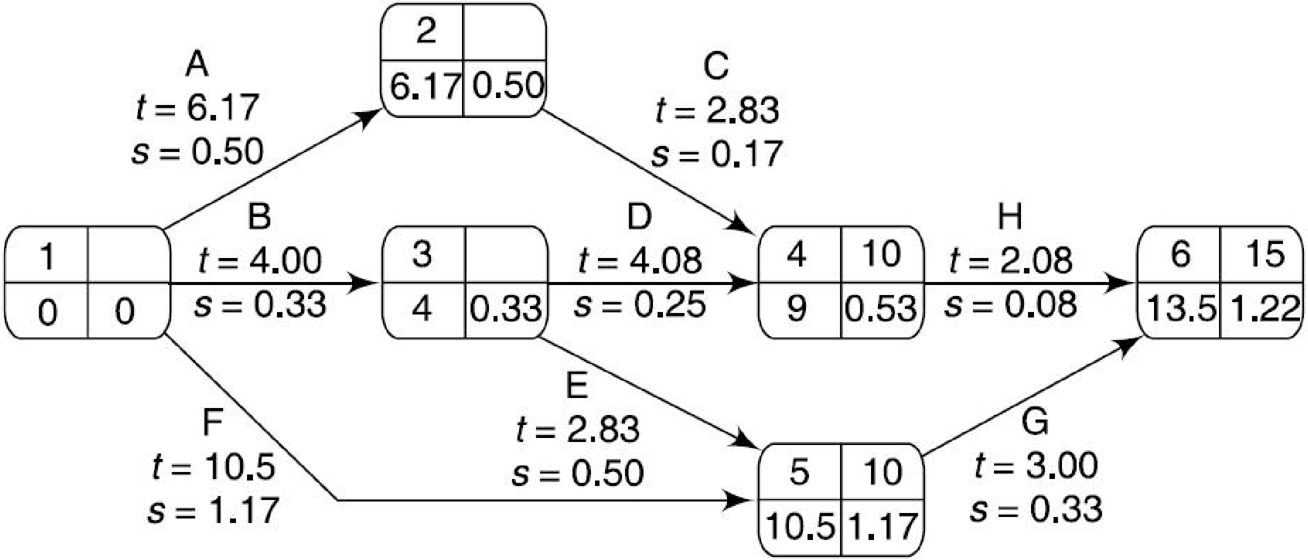
Evaluating risks to the schedule

- Case: PERT Application
  - Activity standard deviations ***s*** of an activity time using the formula: $s = \frac{b-a}{6}$

Activity	Activity durations (weeks)				
	Optimistic (a)	Most likely (m)	Pessimistic (b)	Expected (t_e)	Standard deviation (s)
A	5	6	8	6.17	0.50
B	3	4	5	4.00	0.33
C	2	3	3	2.83	0.17
D	3.5	4	5	4.08	0.25
E	1	3	4	2.83	0.50
F	8	10	15	10.50	1.17
G	2	3	4	3.00	0.33
H	2	2	2.5	2.08	0.08

Evaluating risks to the schedule

- Case: PERT Application
  - The PERT network with three target dates and calculated event standard deviations.



Evaluating risks to the schedule

- Case: PERT Application
  - The PERT technique uses the following three-step method for calculating the probability of meeting or missing a target date.
    - calculate the standard deviation of each project event;
    - calculate the z value for each event that has a target date;
    - convert z values to a probabilities.
  - Step 1: calculate the standard deviation of each project event
    - The standard deviation for event 3 depends solely on that of activity B. The standard deviation for event 3 is therefore 0.33.
    - For event 5 there are two possible paths, B + E or F. The total standard deviation for path B + E is $\sqrt{0.33^2 + 0.50^2} = 0.6$ and that for path F is 1.17; the standard deviation for event 5 is therefore the greater of the two, 1.17.

Evaluating risks to the schedule

- Case: PERT Application
  - Step 2: calculate the *z* value for each event that has a target date
    - The z value is calculated for each node that has a target date. It is equivalent to the number of standard deviations between the node's expected and target dates. It is calculated using the formula:

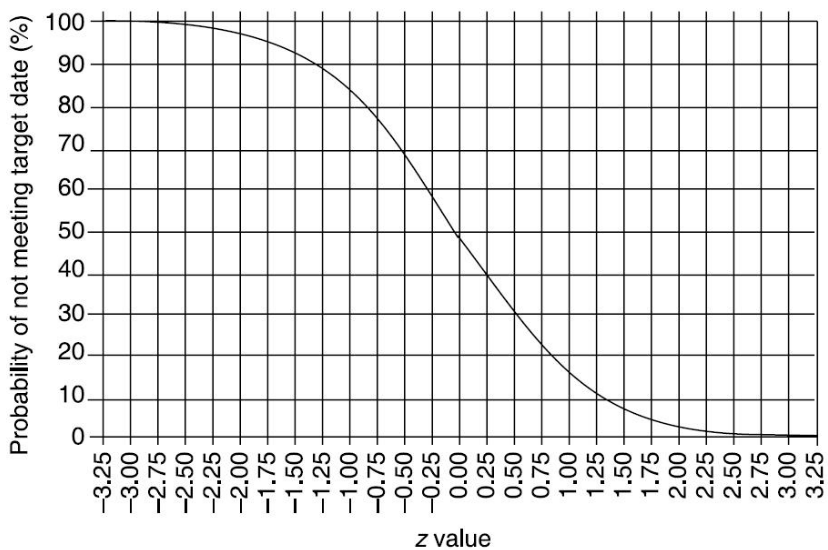
$$z = \frac{T - t_e}{s}$$

where t e is the expected date and T the target date.

    - The *z* value for event 4 is (10 – 9.00)/0.53 = 1.8867.

Evaluating risks to the schedule

- Case: PERT Application
  - Step 3: Converting z values to probabilities
    - A *z* value may be converted to the probability of not meeting the target date by using the graph in the following Figure:



- The *z* value for the project completion (event 6) is 1.22. we can see that this equates to a probability of approximately 11%, that is, there is an 11% risk of not meeting the target date of the end of week 15.

Evaluating risks to the schedule

- The advantages of PERT
  - PERT focuses attention on the uncertainty of forecasting. We can use the technique to calculate the standard deviation for each task and use this to rank them according to their degree of risk. Using this ranking, we can see, for example, that activity F is the one regarding which we have greatest uncertainty, whereas activity C should, in principle, give us relatively little cause for concern.
  - If we use the expected times and standard deviations for forward passes through the network we can, for any event or activity completion, estimate the probability of meeting any set target. In particular, by setting target dates along the critical path, we can focus on those activities posing the greatest risk to the project's schedule.

Summary of Lesson 8

- In this lesson, we have learned how to identify and manage the risks that might affect the success of a project.
- Risk management is concerned with assessing and prioritizing risks and drawing up plans for addressing those risks before they become problems.
- There are many techniques for estimating the effect of risk on the project's activity network and schedule.
- Many of the risks affecting software projects can be reduced by allocating more experienced staff to those activities that are affected.

Assignment of Lesson Eight

- Everyone should submit his/her own biweekly report (from 13 April to 19 April, one page) to the team leader, which should include the ongoing work and the working hour for work in these two weeks. The team leader should summary the progress of project (one page), according to members' report. Please submit the report at 19th April, 2026